

EcoStep - One-Page Outreach Summary

Project lead: Amangeldi Bekaidar | Location: Almaty, Kazakhstan | Status: working prototype, validation stage

What EcoStep Is

EcoStep is a low-cost piezoelectric floor tile prototype that converts footsteps into measurable electrical signals. The current prototype uses a 4 x 4 array of 16 PZT piezoelectric elements inside a layered compression tile.

Why It Matters

High-footfall spaces such as schools, malls, public walkways, and metro stations contain constant human movement. EcoStep investigates whether part of this mechanical energy can be captured, measured, and used for small sensors, indicators, or climate-awareness displays.

Current Evidence	Still Needs Validation
Physical prototype built and photographed	Exact energy per step under known load
16 PZT elements arranged as a 4 x 4 array	Capacitor charging performance
Approximate tile size: 50 cm x 50 cm	Single PZT vs full-array comparison
Rough cost assumption: about USD 75	Exact wiring topology
Arduino logging and Python analysis workflow prepared	Whether demo electronics can be powered only by the tile

Pilot Ask

EcoStep is seeking feedback, measurement support, and controlled demonstration opportunities in Almaty. The first ask is not public deployment. The first ask is a careful conversation about safety, testing, permissions, and whether a small educational pilot could be possible later.

Claim Boundary

Safe wording: working prototype, validation stage, pilot conversation, measurement support. Avoid: deployed product, verified 0.4 J/step, confirmed Almaty Metro partnership, or approved public installation until written proof exists.

Contact

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